INCH-POUND

MIL-DTL-24552B(SH)
8 June 2001

SUPERSEDING
MIL-T-24552A(SH)
3 November 1986

DETAIL SPECIFICATION

TERMINALS AND SEALS, PLUGS AND SWITCH,
CASUALTY POWER SYSTEMS, RECEPTACLE TYPE,
GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers terminals and seals, seals, plugs, and switches that provide emergency electric power connections for casualty power systems. The capacity of these devices is 450-volt alternating current (Vac), 3-phase, 100- and 200-ampere for use on Naval ships, and other characteristics as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not included documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-B-654 - Brazing Alloys, Silver.
QQ-S-365 - Silver Plating, Electrodeposited, General Requirements for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 05Q, Naval Sea Systems Command, 1333 Isaac Hull Ave SE, Washington Navy Yard, Washington D.C. 20376, by using the Standardization Document Improvement Proposal (DD form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5940

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DEPARTMENT OF DEFENSE

MIL-PRF-900	- Rubber Gasket Material, 45 Durometer Hardness.
MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements.
MIL-PRF-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-85T.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-DTL-24552/1	- Terminals and Seals, Plugs and Switch, Casualty Power, Receptacle Type, Back Connected, 200-Ampere, 450-Volt Alternating Current, Three-Phase, Symbol No. 1046.
MIL-DTL-24552/2	- Terminals and Seals, Plugs and Switch, Casualty Power, Receptacle Type, Upper and Lower Riser, 200-Ampere, 450-Volt, Alternating Current, Three-Phase, Symbol No. 1047.
MIL-DTL-24552/3	- Terminals and Seals, Plugs and Switch, Casualty Power, Seal, Symbol No. 1048.
MIL-DTL-24552/4	- Terminals and Seals, Plugs and Switch, Casualty Power, Plug, 100-Ampere, 450-Volt, Alternating Current, Three-Phase, Symbol No. 1049.
MIL-DTL-24552/5	- Terminals and Seals, Plugs and Switch, Casualty Power, Switch, Portable 100-Ampere, 450-Volt, Alternating Current, Three-Phase, Symbol No. 1041.1.
MIL-C-39029	- Contacts, Electrical Connector, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.
FED-STD-H28/2	- Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads - UN and UNR Thread Forms.
FED-STD-595	- Colors Used in Government Procurement.

DEPARTMENT OF DEFENSE

MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1	- Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1344	- Test Methods for Electrical Connectors.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the

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issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 313 - Standard Specification for Stainless Steel Spring Wire. (DoD adopted)
ASTM B 187 - Standard Specification for Copper Bar, Bus Bar, Rod and Shapes.
ASTM B 633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel. (DoD adopted)
ASTM B 695 - Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel.
ASTM D 2240 - Standard Test Method for Rubber Property - Durometer Hardness. (DoD adopted)
ASTM D 5948 - Standard Specification for Molding Compounds, Thermosetting.
ASTM F 1836M - Standard Specification for Stuffing Tubes, Nylon, and Packing Assemblies.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS R 6855 Elastomer, Synthetic, Sheets, Strips, Molded or Extruded Shapes, General Specification.

(Application for copies should be directed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8 - Specification for Filler Metals for Brazing. DoD adopted)

(Application for copies should be addressed to the American Welding Society, 550 NW LeJeune Road, Miami, FL 33126.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Unless otherwise specified (see 3.1), the casualty power terminals, plugs, and seals furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Materials and processes. Materials and processes shall be as specified herein and in 3.1.

3.3.1 Metals. Metals shall be in accordance with MIL-E-917 to provide corrosion resistance. Metals that are not inherently corrosion-resistant shall be processed (treated, plated, or painted) to provide corrosion resistance. Cadmium treatments shall not be used.

3.3.1.1 Selection of metals in direct contact. In order to minimize corrosion attack due to electrolytic action between dissimilar metals in

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contact with each other, metal-to-metal contact shall be limited to those metals which, when coupled, are in accordance with MIL-STD-889.

3.3.2 Nonmetallic materials. Nonmetallic materials shall be nonflammable and nontoxic in accordance with MIL-E-917. Materials containing asbestos shall not be used. PVC materials shall not be used.

3.3.2.1 Thermosetting insulating material. Thermosetting molded insulating parts shall be type MAI-60 in accordance with ASTM D 5948, or one of the following: Fiberite No. 5064, ARMC No. 446 or equal.

3.3.2.2 Gaskets and seals. Material for gaskets and seals shall be in accordance with SAE-AMS-R-6855, class 2, type B or in accordance with MIL-PRF-900.

3.3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Parts.

3.4.1 Contacts. The contact pins and contact sockets shall be of a copper or copper-alloy material that will provide the required electrical characteristics. The contacts shall be silver-plated with not less than 0.0002 inch of silver in accordance with QQ-S-365.

3.4.1.1 Contact pins. Plug-mounted pins shall meet the physical, electrical, and performance requirements as specified in 3.1 and 3.6. The contact-pin and end lugs shall be crimp style and insulated. Lugs shall be appropriately sized to accept American Wire Gauge (AWG) size no. 4 copper conductors applied with standard crimping tools.

3.4.1.2 Contact sockets. The terminal contact sockets shall be sized to accept the contact pins and meet the physical, electrical, and performance requirements as specified in 3.1 and 3.6. Socket contact springs, if required for mechanical strength, shall be corrosion-resistant steel wire in accordance with ASTM A 313. The spring shall not form a part of the electrical circuit.

3.4.2 Terminal lugs. Terminal assembly lugs shall be of a copper or copper-alloy material, crimp style, and insulated. Lugs shall be sized for attachment to 0.250-inch diameter studs and to accept conductors size no. 2 wire. Lugs shall be formed to a 45-degree angle to aid in installation of conductors.

3.4.3 Hardware. Hardware (bolts, nuts, screws, washers, and miscellaneous hardware) shall be of a good commercial grade material compatible with that of the basic unit. Steel (corrosion-resisting steel excepted) hardware shall be zinc-plated in accordance with ASTM B 633 or ASTM B 695.

3.5 Design and construction. Terminal, plugs, seals, and switches shall be of the design, construction, and physical dimensions specified (see 3.1).

3.5.1 Mounting. The size and location of mounting holes shall be as specified (see 3.1).

3.5.2 Housings. Housings for terminals, plugs, covers, and switch enclosure shall be of a glass-fiber, reinforced molded thermosetting insulating material. The color shall be red (color no. 11105 or 11136 in accordance with FED-STD-595).

3.5.2.1 Housing - Terminal. The terminal housing shall have a

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removable access cover and a cavity of sufficient size to enable ease of making internal electrical connections and shall have 3-phase outlets containing bus, assembled socket contacts, and protective covers. The socket contacts shall be removable, rigidly mounted, and shall allow for a radial misalignment of plus or minus 0.020 inch. The outlets shall be polarized to prevent cross-phasing and shall contain threaded inserts for retaining plugs and covers. Removal of plugs and protective covers shall not cause the socket contacts, connecting bus, or wire to become loose.

3.5.2.2 Housing - Plug. The plug housing shall protrude beyond and afford protection for the contacts. The plugs shall have male contact pins, retaining devices that mate with the terminal housing threaded inserts, and shall be polarized to prevent cross-phase engagement. To facilitate self-alignment of mating contacts, the contact pins shall be retained so that the individual contact shall have an overall sideplay of 0.002 to 0.006 inch. To prevent accumulation of condensation or moisture while connected to the terminals, vent holes shall be utilized.

3.5.2.3 Protective covers. A protective cover shall provide a spray-tight seal at the terminal or switch interface when fully engaged and retained with devices specified in 3.5.3. Vent holes shall be incorporated into the cover to allow condensation and moisture to drain. The protective cover shall be held captive to the terminal housing or switch enclosure by means of a flexible, plastic-coated, stainless steel cable. Phase identification (A, B, and C) shall not be marked on the protective covers.

3.5.3 Retaining devices. The protective cover shall contain captive retaining devices. The retaining devices shall provide positive closure and release without the use of tools. The retaining devices shall not be used as an aid for mating or separating plugs or covers to and from terminals.

3.5.4 Plug engaging and separating. Lubricants, tools, retaining, or coupling devices shall not be required for engaging or separating a plug from the receptacle.

3.5.5 Cable entry boot - Plug. A removable flame-retardant, polyetherurethane cable entry boot shall be provided with each plug. The shore A hardness change of the boot shall be not greater than 10 after being subjected to the humidity test of 4.7.8. The boot shall be held in place with a stiffener plate to prevent water leakage into the cable end of the plug housing.

3.5.6 Cable entry seal - Upper and lower terminal riser. When specified (see 3.1), a nylon stuffing tube shall be furnished with each upper and lower terminal riser.

3.5.7 Screw threads for fastening devices. Screw threads for all threaded fastening devices shall be in accordance with FED-STD-H28. Unless otherwise specified (see 3.1), the threads shall be right-hand, coarse-thread series, unified form, class 2A or 2B in accordance with FED-STD-H28/2. Unless otherwise specified (see 3.1), when a screw mates with a plastic part, a threaded metal insert, bushing type or helical type, shall be molded therein. The bushing type is the preferred type.

3.5.7.1 Thread locking. Nuts, bolts, studs, and screws used for electrical connections shall be secured by lockwashers. Bolts, nuts, and screws used for mechanical connections shall be secured by locking devices, as required, to ensure compliance with the requirements specified (see 3.1).

3.5.8 Brazing. Brazing alloys for electrical connections shall be in accordance with AWS A5.8 or QQ-B-654.

3.5.9 Electrical rating. The electrical rating of the terminals shall be 200-ampere per phase, 450-Vac, 3-phase. Rating of plugs and switches shall be 100-ampere per phase, 450-Vac, 3-phase.

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3.5.10 Phase identification. Phase identification (A, B, and C) shall be permanently and legibly marked on the plug and terminal housings in characters at least 0.12 inch high. Phase orientation shall be A, B, and C, respectively, front to rear and right to left (as viewed from front) as specified (see 3.1).

3.5.11 Electrical creepage and clearance distances. Clearance distance in air between energized parts or between an energized part and ground shall be not less than 0.500 inch. Creepage distance between energized parts or between an energized part and ground shall be not less than 0.875 inch.

3.5.12 Bus bar. Terminal and switch bus bars shall be copper, hard temper in accordance with ASTM B 187. The bus shall be solid and shall have a cross sectional area of not less than 0.0625 square inch (in^2).

3.5.12.1 Plating. Bus bars shall be silver-plated in areas of current-carrying contact with bus ties, contacts, and terminals. The plating shall be in accordance with QQ-S-365, with the exception that the silver plate shall be not less than 0.0002 inch thick. Silvering a 0.125-inch wide band on the surface area surrounding each hole used for retaining an electrical-carrying contact will be acceptable; however, the entire bus may be silver-surfaced at the discretion of the manufacturer. Threaded surfaces used as electrical contact surfaces shall have silver thickness of at least 0.0002 inch.

3.6 Performance characteristics.

3.6.1 Dielectric withstanding voltage. There shall be no arcing, breakdown, corona (audible or visible), or punctured insulation when plugs and terminals or switches are tested as specified in 4.7.1.

3.6.2 Insulation resistance. With plugs and terminals or switches unmated, the insulation resistance between electrical conductors and between electrical conductors and ground shall be not less than the values indicated, when tested as specified in 4.7.2:

a. 500 megohms - Terminals and plugs.
b. 1 megohm - Switches.

3.6.3 Contact engaging and separating forces. The required engaging force shall be not greater than 100 ounces, and the separation force shall be not less than 10 ounces when individual contacts are tested as specified in 4.7.3.

3.6.4 Plug engaging and separating forces. The required force for engaging or separating a plug from a receptacle (see 3.5.4) shall be not greater than 20 pounds when tested as specified in 4.7.4.

3.6.5 Cable pull-out. Cables shall not pull out of their respective plugs when a 360-pound load is applied as specified in 4.7.5.

3.6.6 Ambient temperature extremes. When tested as specified in 4.7.6, plugs and terminals shall withstand ambient temperature extremes of minus 20 degrees Celsius (^{o}C) to plus 125°C without showing evidence of mechanical damage as follows:

a. Cracking or delamination of finish.
b. Mechanical displacement or rupture of conductors or insulating materials.

Following the test, terminals and plugs shall pass tests specified in 4.7.1 and 4.7.2, respectively. Bulkhead terminals shall also pass the test specified in 4.7.16.

3.6.7 Contact resistance. Terminals which have not undergone the salt

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spray test shall exhibit a voltage drop not greater than 40 millivolt (mV) when 100 amperes are passed through each connector when tested as specified in 4.7.7. Following the salt spray test, voltage drop shall be not greater than 50 mV.

3.6.8 Humidity. Mated plugs and terminals shall pass the tests as specified in 4.7.1 and 4.7.2 following a 4-hour drying period when tested as specified in 4.7.8.

3.6.9 Fluid immersion. The plugs shall be properly mated with the terminal (see 3.5.4) after having been immersed in fuel oil or turbine fuel as specified in 4.7.9.

3.6.10 Electrical load. A mated plug and terminal or plug and switch shall operate for 4 hours without evidence of electrical or mechanical damage when subjected to the load specified in 4.7.10. Following this test, the items shall pass the test specified in 4.7.1 and 4.7.2, respectively.

3.6.11 Endurance (durability). The plugs and receptacles shall be serviceable and perform their function by passing the test specified in 4.7.7 after completion of the test specified in 4.7.11.

3.6.12 High-impact (HI) shock. Terminals with plugs mated shall be in accordance with HI shock requirement, type A, grade A, class I, lightweight of MIL-S-901 (see 4.7.12). The items being tested shall be considered as failing the test if one or more of the following occurs:

a. There is evidence of contact of live parts with each other.
b. Structural parts are damaged or loosened.
c. Functional parts are damaged or loosened.
d. The housing is cracked or warped.
e. The circuit breaker trips (switches only).
f. Failure to pass the test specified in 4.7.1 or the test specified in 4.7.2 upon completion of the high-impact shock test.

3.6.13 Mechanical abuse (impact). There shall be no evidence of mechanical damage or electrical failure that would prevent the plug from performing its intended function when plugs are tested as specified in 4.7.13.

3.6.14 Vibration. Terminals, switches, and plugs shall conform to the type I vibration requirements of MIL-STD-167-1 when tested as specified in 4.7.14. The items being tested shall be considered as failing the test if they fail to perform their principal functions during or after testing or if one or more of the deficiencies specified in 3.6.12 (a) through 3.6.12(e) occurs. Following the vibration tests, the items shall pass the tests specified in 4.7.1 and 4.7.2.

3.6.15 Salt spray. There shall be no evidence of excessive corrosion when terminals, switches, or plugs are tested as specified in 4.7.15. Excessive corrosion is defined as that which interferes with electrical or mechanical performance, or in the case of plated metals, corrosion which has migrated through the plating and attacked the base metal. After the test, the plugs and terminal shall pass the test specified in 4.7.7.

3.6.16 Watertightness. Back-connected terminals with seals shall withstand a water gauge pressure test as specified in 4.7.16. Upper and lower riser terminals and switch enclosures shall meet the watertight requirements as specified in 4.7.17.

3.7 Finish.

3.7.1 Molded parts. Molded parts shall have surfaces which are not warped, cracked, chipped, or blistered either initially or as a result of tests specified herein.

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3.8 Identification plate. Unless otherwise specified (see 6.2), the following information shall be permanently installed on the terminals, plugs, and switch enclosures by molding on to the items or by installing identification and information plates:

a. Military part number.
b. Symbol number.
c. Manufacturer's name, logo, Commercial and Government Entity (CAGE) code.
d. Contract number, optional, or when required by Contracting Officer.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of the examination and tests as specified in table I in the order shown. Qualification inspection shall be conducted at a laboratory satisfactory to the Government (see 6.3) on sample units produced with equipment and procedure normally used in production (see 4.4).

4.2.1 Extension of qualification. Except for the cable entrance, qualification approval of back-connected terminals may be extended to cover the upper and lower riser terminals, provided they are of the same design and construction and that qualification testing for high-impact shock and vibration was accomplished without the use of a bulkhead seal.

TABLE I. Qualification and inspection.

Examination/test	Sample 1	Sample 2	Requirement	Test method
General examination	X	X	3.1, 3.3, 3.4, 3.5, 3.7, and 3.8	4.6.1
Dielectric withstanding voltage		X	3.6.1	4.7.1
Insulation resistance	X	X	3.6.2	4.7.2
Contact engaging and separating force	X	X	3.6.3	4.7.3
Plug engaging and separating force	X	X	3.6.4	4.7.4
Cable pull-out 1/	X	X	3.6.5	4.7.5
Ambient temperature extremes 2/	X	X	3.6.6	4.7.6
Contact resistance	X		3.6.7	4.7.7
Humidity 2/		X	3.6.8	4.7.8
Fluid immersion 1/		X	3.6.9	4.7.9
Electrical load 2/		X	3.6.10	4.7.10
Durability		X	3.6.11	4.7.11
High impact shock	X		3.6.12	4.7.12
Mechanical abuse 1/	X		3.6.13	4.7.13
Vibration	X		3.6.14	4.7.14
Salt spray	X		3.6.15	4.7.15
Watertightness 3/	X		3.6.16	4.7.16
Watertightness 4/	X	X	3.6.16	4.7.17

1/ Plugs only.
2/ Not applicable for switches.
3/ Bulkhead terminals only (see 6.1).

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4/ Upper and lower riser terminals and switch enclosure.

4.3 Conformance inspection.

4.3.1 Inspection lot. An inspection lot shall consist of all plugs, terminals, switches, or seals of the same part number produced under essentially the same conditions and offered for delivery at one time.

4.3.2 Group A inspection. Group A inspection shall consist of the inspections as specified in table II. Classification of defects shall be as specified in table III.

TABLE II. Group A inspection.

Inspection	Requirement	Test method
Visual and dimensional examination	3.1, 3.3, 3.4, 3.5, 3.7, and 3.8	4.6.1

TABLE III. Classification of defects.

Categories	Characteristics
Critical:	
None	
Major:	
101	Inability to mate with counterpart
102	Incorrect contact construction
103	Incorrect dimensions which affect mating
104	Burrs capable of cutting
105	Incorrect marking
106	Cracked, loose, or displaced parts
107	Piece part missing or not as specified
108	Material incorrect
109	Dimensions incorrect
Minor:	
201	Poor exterior finish
202	Molding flash
203	Workmanship not up to standards

4.3.3 Group B inspection. Group B inspection shall consist of the inspections as specified in table IV with samples selected from inspection lots that have passed group A inspection.

TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Dielectric withstanding voltage	3.6.1	4.7.1
Insulation resistance	3.6.2	4.7.2
Contact engaging and separating force	3.6.3	4.7.3
Plug engaging and separating force	3.6.4	4.7.4
Ambient temperature extremes	3.6.6	4.7.6
Contact resistance	3.6.7	4.7.7
Watertightness	3.6.16	4.7.16

4.3.4 Noncompliance. If a sample fails to pass group B inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection

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activity of such failure.

4.4 Sample size. The contractor shall submit two sample units of the type for which qualification is desired (see 4.2.1). A sample unit shall consist of: One terminal and three plugs for riser types; one terminal, three plugs, and one seal for back connected types; and one switch and two plugs for switch assemblies.

4.5 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions as specified in MIL-STD-202. Except for the tests specified in 4.7.1 and 4.7.7, the plugs and terminals shall be terminated using cable and wire as noted in 6.5.

4.6 Examination.

4.6.1 Visual and dimensional examination. Plugs, terminals, switches, and seals shall be visually examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are as specified in the applicable requirements (see 3.1, 3.3, 3.4, 3.5, 3.7, and 3.8).

4.7 Test methods.

4.7.1 Dielectric withstanding voltage. Plugs and terminals or switches shall be subjected to a dielectric withstanding voltage test in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Nature of potential - 60 hertz (Hz) root mean square (rms) ac and shall approximate a true sine wave.
b. Magnitude of test voltage - 2500 volts rms (terminals and plugs), 2000 volts rms (switches).
c. Points of application - When testing materials, the potential shall be applied between each phase and the metal mounting plate and between each phase and all adjacent phases. When testing plugs, the potential shall be applied between each phase and between each phase and any metallic parts.
d. Rejection criteria - Failure to be as specified in 3.6.1.

4.7.2 Insulation resistance. Insulation resistance shall be measured in accordance with method 302 of MIL-STD-202. The test voltage (500 volts direct current (Vdc)) shall be applied between electrical conductors and between electrical conductors and ground. When switches are present, they shall be in the energized (closed) position. A resistance of less than the values specified in 3.6.2 shall be cause for rejection.

4.7.3 Contact engaging and separating forces. Individual terminal sockets shall be tested for initial contact engaging and separating force in accordance with MIL-C-39029 for size 4 contacts (see 3.6.3).

4.7.4 Plug engaging and separating forces. Mated plugs and terminals or switches shall be tested in accordance with method 2013 of MIL-STD-1344. Exceeding the force specified in 3.6.4 shall be cause for rejection.

4.7.5 Cable pull-out. Three plugs shall be individually placed in a tensile testing machine. Tests shall be made at room temperature and with a head speed of ½ inch per minute, until load specified in 3.6.5 has been applied. Evidence of discontinuity or relative movement between any conductor and its termination shall be cause for rejection.

4.7.6 Ambient temperature extremes. A terminal with one cabled plug and two covers mated shall be subjected to a high and low temperature test in accordance with method 501.2, procedure II and method 502.2, procedure II of MIL-STD-810, except that the high temperature shall be 125°C and the low temperature shall be minus 20°C (see 3.6.6). The samples shall then be subjected to the dielectric withstanding voltage test of 4.7.1 and the

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insulation resistance test of 4.7.2. Failure of these tests or evidence of mechanical damage as specified in 3.6.6 shall be cause for rejection.

4.7.7 Contact resistance. Mated terminals and test plugs shall be connected to a power supply and circuitry as needed to pass 100 amperes through each contact during the test. Test leads shall be not less than 5 feet. A potential drop greater than that specified in 3.6.7, when measured at the extreme ends of the contacts (see figure 1), shall be cause for rejection.

4.7.8 Humidity. Mated plugs and terminal shall be subjected to a moisture resistance test in accordance with method 103, condition C of MIL-STD-202. Upon completion and after a 4-hour drying period, the plugs and terminal shall be subjected to the tests of 4.7.1 and 4.7.2. Boot (plug) shall be subjected to a hardness test in accordance with ASTM D 2240 prior to and following the humidity test. Change in shore A hardness greater than specified in 3.5.5 shall be cause for rejection.

4.7.9 Fluid immersion. One plug shall be immersed fully for a period of 20 hours at 60 $\pm$ 20C in fuel oil in accordance with MIL-F-16884 and one in turbine fuel in accordance with MIL-PRF-5624. Plugs shall be wiped free of the fluids and connected to a terminal to determine proper mating as specified in 3.6.9.

4.7.10 Electrical load. Two plugs shall be mated to a terminal as shown on figure 2, and each plug lead shall be connected to a 100-ampere load. A power supply capable of supplying 200 amperes through all phases simultaneously shall be connected to the terminal bus bar. The test current shall be applied for 4 hours. Upon completion of the test, the terminal and plug shall be subjected to the tests of 4.7.1 and 4.7.2.

4.7.11 Endurance (durability). Plugs and receptacles shall be subjected to the durability test in accordance with method 2016 of MIL-STD-1344 to meet the requirements specified in 3.6.11. The following details shall apply:

a. Number of mating and unmating cycles: 500.
b. Cycle rate: 5 cycles per minute minimum.
c. Plug retention devices may be removed for this test.
d. Rejection criteria: Evidence of damage in accordance with method 2016 of MIL-STD-1344 or failure to be as specified in 3.6.7.

4.7.12 High-impact shock. The terminals or switch, with plugs mated as indicated, shall be energized and subjected to high-impact shock tests in accordance with MIL-S-901 for grade A, class I, lightweight, type A equipment. The manner on mounting shall simulate the actual mounting on shipboard. Connections shall be as follows:

a. Bulkhead terminal (see figure 3) - Three plugs, each with 6- to 10-foot lengths of unsupported cable attached, installed to one side. One plug, with a 6- to 10-foot length of unsupported cable attached, and two covers installed on the opposite side.
b. Terminal risers - Two plugs, each with a 6- to 10-foot length of unsupported cable attached, and two covers installed on the opposite side.
c. Switch - Two plugs, each with a 6- to 10-foot length of unsupported cable attached.

The sample being tested shall not be reconditioned or adjusted during the testing. Following the test, the samples shall be subjected to tests of 4.7.1 and 4.7.2. The items tested shall be considered as failing the test if damage specified in 3.6.12 is evident, or if they fail to pass the tests specified in 4.7.1 and 4.7.2.

4.7.13 Mechanical abuse (impact). A cabled plug shall be subjected to the test in accordance with method 2015 of MIL-STD-1344 to verify compliance

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with the requirements specified in 3.6.13. The following details and exceptions shall apply:

a. Service class: Severe.
b. Drop height: As specified for severe service class.
c. Drop surface: 0.050-inch minimum steel plate, or 4-inch maximum concrete slab.
d. Rejection criteria: Evidence of damage in accordance with method 2015 of MIL-STD-1344 or failure to pass the dielectric withstanding voltage test specified in 4.7.1 or the insulation resistance test specified in 4.7.2.

4.7.14 Vibration. The terminals or switches with cabled plugs mated shall be energized and shall, while mounted to simulate a shipboard installed condition, be subjected to the type I vibration test in accordance with MIL-STD-167-1. Acceptability criteria shall be as specified in 3.6.14. Following this test, the samples shall be subjected to the tests specified in 4.7.1 and 4.7.2.

4.7.15 Salt spray. Unmated plugs and terminals, with covers secured, shall be subjected to the salt spray test, method 101, condition A in accordance with MIL-STD-202. The salt solution shall be 5 percent. The equipment shall not be energized during the test. Upon completion, the plugs shall be examined to determine conformance with 3.6.15 and shall be subjected to the test specified in 4.7.7. (Prior to making electrical measurements, the test samples shall be cleaned in accordance with MIL-STD-202.)

4.7.16 Watertightness (bulkhead terminals and seals). A back-connected terminal with seal shall be mounted to simulate one side of a bulkhead terminal assembly as shown on figure 3. The terminal, with covers removed and with wiring connected to the terminal through the seal and a simulated bulkhead, shall be subjected to a water gauge pressure of 6.5 pounds per square inch (lb/in^2) for 2 hours to determine its watertightness as specified in 3.6.16. The following details and exceptions shall apply:

a. Rejection criteria: Evidence of water leakage through the bulkhead.
b. Water seepage through the conductor shall not be considered failure of the test.

4.7.17 Watertightness (terminals and switch enclosures). A terminal with protective covers installed and cable entrance blanked off, and a switch enclosure with protective covers installed shall be subjected to the watertightness test in accordance with MIL-STD-108 to meet the requirements specified in 3.6.16. Entrance of water into the interior of the terminal or switch enclosure shall be cause for rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department or Defense Agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Terminals, plugs, switches, and seals covered by

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this specification are intended for use with the casualty power system on Naval ships. These items are part of the casualty power interconnect system that is installed on board ships to provide temporary transmission of power to span damaged areas.

a. Upper and lower riser terminal (see 6.5 for cable information). Used to provide transmission of power between decks.
b. Back-connected terminal (for use with interconnecting hook-up type wire, AWG size no. 2). Used to provide transmission of power through non-watertight bulkheads and from switchboards or equipment enclosures.
c. Bulkhead terminal, watertight (for use with two back-connected terminals and seals, and interconnecting wire, AWG size no. 2). Used to provide transmission of power through watertight bulkheads. The bulkhead terminal installation consists of two back-connected terminals and two rubber seals in accordance with MIL-DTL-24552/3 to prevent water leakage through the bulkhead without the aid of mated plugs or covers. Bulkhead terminal is mounted as shown on figure 3.
d. Switches are used with connectorized cable, AWG size no. 4 (see 6.6). The cables provide a means of supplying and interrupting power to loads that may require intermittent power from the casualty power systems.
e. Plugs are used with the terminals and switches and are furnished unterminated. When completely assembled, the plugs form part of the cable assembly.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Title, number, and date of the applicable specification sheet, and the part number (see 1.1, 3.1, and 6.4).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. When first article inspection is required (see 3.1).
e. Marking if other than as specified (see 3.8).
f. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-24552 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Sea Systems Command, SEA 05Q, Department of the Navy, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

6.4 Part or identifying number (PIN). The part numbers consist of the letter “M” followed by the general specification number, specification sheet number, and a sequentially assigned dash number (see 3.1 and 6.2).

Example: M24552/1 - XX

M	24552	/1	-XX
M prefix	General specification number	Applicable specification sheet number	Dash number (see specification sheets for dash numbers)

6.5 Cables and wire. The plugs and terminals should be capable of being used with the following types of cable and wire:

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a. Plugs - Cable type should be 3-conductor, 600-volt rated, AWG size no. 4, oil-resistant, non-watertight, for flexing service, with a low-smoke, cross-linked polyolefin jacketing.
b. Terminals - Cable type should be 3-conductor, 1000-volt rated, AWG size no. 1 with a low-smoke, cross-linked polyolefin jacket in accordance with IEEE 45. Bulkhead terminals may be connected with hook-up wire, single-conductor, AWG size no. 2.

6.6 Subject term (key word) listing.

Electrical power connections

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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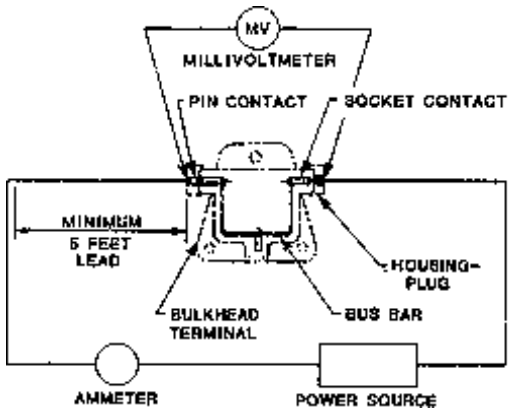


FIGURE 1. Socket contact bus assembly.

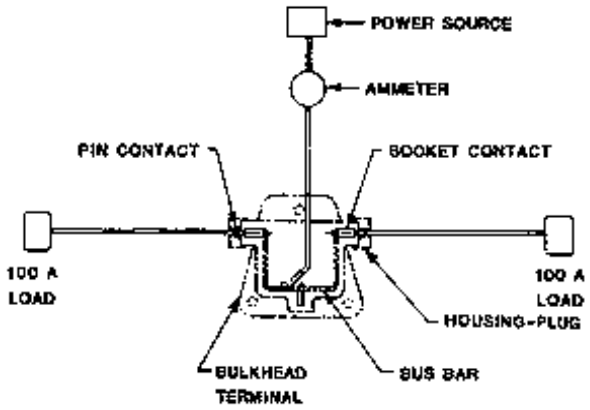


FIGURE 2. Electrical load test configuration.

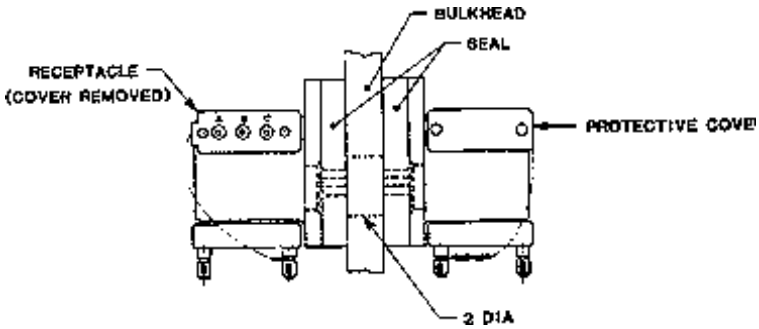


FIGURE 3. Typical bulkhead terminal installation.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7 and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-24552B	2. DOCUMENT DATE *(YYYYMMDD)* 20010608

3. DOCUMENT TITLE TERMINALS AND SEALS, PLUGS AND SWITCH, CASUALTY POWER SYSTEMS, RECEPTACLE TYPE, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(if applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME Ruth Butler	b. TELEPHONE *(Include Area Code)* (1) Commercial (202) 781-3726 (2) DSN 326-3726
c. ADDRESS *(Include Zip Code)* Commander, Naval Sea Systems Command ATTN: SEA 05Q, 1333 Isaac Hull Ave SE, Washington Navy Yard, Washington, D.C. 20376	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533, Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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